

BANGOR, ME, USA

WMO: 726070

Lat: 44.798N

Lon: 68.819W

Elev: 148

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14606

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-1.6	-17.3	2.2	-4.1	-12.3	2.8	0.5	27.1	24.7	24.2	22.1	5.0	310	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	87.5	71.0	84.0	69.2	81.1	67.2	73.2	83.0	71.3	80.5	69.6	77.4	9.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	111.3	77.6	68.3	104.6	75.4	66.7	98.7	73.7	36.8	83.0	35.2	80.5	33.7	77.8	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	19.8	17.8	DB	-15.3	92.9	6.1	2.3	-19.7	94.5	-23.3	95.8	-26.7	97.1	-31.1	98.7	
			WB	-15.7	76.6	6.0	1.8	-20.0	77.9	-23.5	78.9	-26.9	80.0	-31.3	81.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	18.6	21.3	30.6	42.8	54.4	63.5	69.6	68.0	60.3	48.5	37.1	26.0	(6)
	DBStd	19.35	11.71	10.62	9.50	7.04	6.50	6.39	4.97	4.94	6.91	7.31	8.56	10.05	(7)
	HDD50	3901	973	803	605	233	26	1	0	0	4	117	394	745	(8)
	HDD65	7601	1438	1223	1067	665	336	102	15	24	173	512	836	1210	(9)
	CDD50	2149	0	0	3	18	162	406	608	559	314	71	8	0	(10)
	CDD65	373	0	0	0	0	7	57	157	118	33	1	0	0	(11)
	CDH74	3174	0	0	6	7	138	566	1263	923	260	11	0	0	(12)
	CDH80	862	0	0	1	1	36	160	356	238	70	0	0	0	(13)
(14) Wind	WSAvg	7.6	7.6	8.0	9.1	9.1	8.0	7.2	6.6	6.3	6.7	7.4	7.5	7.4	(14)
(15) Precipitation	PrecAvg	39.4	3.0	2.4	3.5	3.4	3.4	3.4	3.1	3.0	3.4	3.5	3.7	3.6	(15)
	PrecMax	65.2	7.6	5.2	6.7	11.0	7.6	7.0	7.2	6.6	6.9	9.0	11.6	9.2	(16)
	PrecMin	23.2	0.5	0.6	0.6	0.6	0.6	1.0	0.9	0.7	1.1	1.0	1.0	1.3	(17)
	PrecStd	8.3	1.9	1.1	1.3	2.0	1.9	1.5	1.7	1.5	1.6	1.8	2.0	2.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	50.1	63.5	72.7	85.0	90.1	90.5	90.4	87.9	75.2	62.7	54.4	(19)
		MCWB	50.5	45.8	51.1	55.3	65.1	71.5	72.7	72.1	71.6	61.8	57.4	52.0	(20)
	2%	DB	45.5	44.7	52.3	64.7	77.5	84.5	87.3	85.8	80.8	68.3	58.7	49.0	(21)
		MCWB	42.8	40.7	43.5	50.5	61.0	68.7	70.9	70.6	67.2	58.9	54.3	46.4	(22)
	5%	DB	40.0	40.4	47.6	60.0	72.2	80.4	84.0	82.1	76.0	64.3	55.0	45.0	(23)
		MCWB	37.0	36.7	41.0	47.9	57.8	65.8	69.6	68.3	65.0	57.6	50.6	41.8	(24)
	10%	DB	35.6	37.1	43.4	55.4	67.7	76.5	81.1	79.2	72.3	61.0	51.1	40.1	(25)
		MCWB	32.9	33.8	38.3	45.5	55.5	63.5	67.8	66.4	62.8	55.4	47.4	37.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	46.3	53.0	58.9	68.1	74.3	76.3	75.0	72.9	65.4	59.6	52.6	(27)
		MCDB	51.1	47.9	59.5	67.4	79.5	86.0	86.2	85.4	84.4	68.8	61.8	54.2	(28)
	2%	WB	43.9	41.5	46.3	54.3	63.5	71.1	73.3	72.7	70.0	62.0	55.7	47.2	(29)
		MCDB	45.2	44.0	50.0	60.3	74.5	81.4	82.6	81.9	77.4	65.7	57.8	48.6	(30)
	5%	WB	37.3	37.4	41.9	50.6	60.3	68.6	71.5	70.7	67.6	59.2	51.6	42.3	(31)
		MCDB	39.5	39.8	46.3	56.3	68.9	77.0	80.6	78.7	72.6	63.8	53.8	44.4	(32)
	10%	WB	33.1	34.4	38.5	47.4	57.5	66.0	69.7	68.9	65.1	56.0	47.4	37.4	(33)
		MCDB	35.0	36.8	42.5	54.3	64.9	73.3	77.6	75.9	69.9	60.1	50.7	39.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	19.1	17.7	19.2	20.4	20.1	20.0	20.6	20.4	18.3	15.7	15.9	(35)
		MCDBR	20.5	19.2	23.1	28.3	29.3	26.4	23.8	24.3	24.2	22.9	19.9	19.6	(36)
	5% WB	MCWBR	19.7	16.2	15.9	16.6	15.3	13.4	10.8	11.2	13.0	15.0	16.5	17.9	(37)
		MCWBR	20.9	17.6	20.4	23.1	24.5	22.8	20.5	20.5	20.1	19.5	17.3	19.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.278	0.289	0.316	0.347	0.371	0.411	0.425	0.400	0.362	0.327	0.307	0.286	(40)
		taud	2.453	2.408	2.450	2.424	2.397	2.286	2.265	2.331	2.424	2.531	2.551	2.502	(41)
	Ebn at Noon		270	287	292	291	285	272	267	270	272	270	258	256	(42)
		Edh at Noon	22	28	31	35	36	41	42	37	31	25	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		504	779	1160	1433	1616	1751	1810	1622	1314	824	511	386	(44)
	RadStd		48	70	74	151	200	138	143	95	81	76	47	36	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A

Nomenclature: See separate page